

Research Highlights

Manuscript Title: A High-Resolution Finite-Difference Benchmark for Steady Continuation and Hopf Bifurcation Dynamics up to Extreme Reynolds Numbers ($Re = 100,000$) in 2D Lid-Driven Cavity Flows

Author: Kartikey Singh

Affiliation: Department of Mathematics, University of Delhi, Delhi, 110007, India

Target Journal: *Computers & Fluids* (Elsevier)

- Exact $\mathcal{O}(N^2 \log N)$ DST-I Poisson solver eliminates iterative residual drift.
- Pure central differencing retains zero numerical viscosity for $Re \leq 50,000$.
- Interior hybrid scheme stabilizes corner singularity at $Re = 100,000$.
- Core vorticity profiles show qualitative alignment with Batchelor theorem.
- Unsteady marching on 513×513 grids resolves shedding limit-cycle dynamics.